

Initial Setup

- **Tape:** 1101a11
- **Head:** at first symbol (1)
- **State:** I (Go to beginning of OP2)

Step-by-Step Execution

Cycle 1

1. **State I $\rightarrow$ J $\rightarrow$ K:** Move right until reaching OP2 start (a $\rightarrow$ 1).

Tape: 1101a11

Head: on last 1 of OP2

State: K

2. **State L:** Subtract 1 from OP2 (change last 1 to 0).

Tape: 1101a10

Head: on modified 0

State: L

3. **State M:** Move left to OP1 end.

Tape: 1101a10

Head: on last 1 of OP1

State: M

4. **State N:** Add 1 to OP1 (binary increment).

OP1: 1101 $\rightarrow$ 1110

Tape: 1110a10

Head: on modified 0

State: N

Return to **I**.

Cycle 2

1. **State I → J → K:** Move to OP2 end.

Tape: 1110a10

Head: on last 0

State: K

2. **State L:** Subtract 1 from OP2 (change 0 → 1? No, borrow logic: find previous 1).

OP2: 10 → 01

Tape: 1110a01

Head: on modified 1

State: L

3. **State M:** Move to OP1 end.

Tape: 1110a01

Head: on last 0 of OP1

State: M

4. **State N:** Add 1 to OP1.

OP1: 1110 → 1111

Tape: 1111a01

Head: on modified 1

State: N

Return to **I**.

Cycle 3

1. **State I → J → K:** Move to OP2 end.

Tape: 1111a01

Head: on last 1

State: K

2. **State L:** Subtract 1 from OP2.

OP2: 01 $\rightarrow$ 00

Tape: 1111a00

Head: on modified 0

State: L

3. **State M:** Move to OP1 end.

Tape: 1111a00

Head: on last 1 of OP1

State: M

4. **State N:** Add 1 to OP1.

OP1: 1111 $\rightarrow$ 10000 (carry overflow)

Tape: 10000a00

Head: on modified 0

State: N

Return to **I**.

Check OP2 Zero

- OP2 = 00 $\rightarrow$ zero detected.
- **State O:** Erase OP2 and a.

Tape: 10000

Head: on first 1

State: O

State P $\rightarrow$ F

- Move head to the start of OP1.
- Halt.

Final Tape

10000

(16 in binary, correct sum of 13 + 3)

Summary Table

Step-by-Step Execution of TM for Binary Addition (1101 + 11)

The Turing Machine described in slides 17–19 performs binary addition by:

- Incrementing OP1 and decrementing OP2 until OP2 becomes zero.
- Erasing the separator `a` and OP2 after completion.
- Returning the head to the start of OP1 and halting.

For input `1101a11` (13 + 3), the expected output is `10000` (16 in binary).

Step Tape Head Pos State

0	1101a11	1	I
1	1101a10	last 0	L
2	1110a10	last 0	N
3	1110a01	last 1	L
4	1111a01	last 1	N
5	1111a00	last 0	L
6	10000a00	last 0	N
7	10000	first 1	F

Final Output

10000

Cycle 1

- Move right to OP2 start (State I $\rightarrow$ J $\rightarrow$ K).
- Subtract 1 from OP2: `1101a11` $\rightarrow$ `1101a10` (State L).
- Add 1 to OP1: `1101` $\rightarrow$ `1110` $\rightarrow$ Tape: `1110a10` (State N).

Cycle 2

- Move to OP2 end.
- Borrow logic: OP2 changes `10` $\rightarrow$ `01` $\rightarrow$ Tape: `1110a01` (State L).
- Add 1 to OP1: `1110` $\rightarrow$ `1111` $\rightarrow$ Tape: `1111a01` (State N).

Cycle 3

- Subtract 1 from OP2: `01` $\rightarrow$ `00` $\rightarrow$ Tape: `1111a00` (State L).
- Add 1 to OP1 with carry overflow: `1111` $\rightarrow$ `10000` $\rightarrow$ Tape: `10000a00` (State N).

Final Steps

- OP2 is zero $\rightarrow$ erase `a` and OP2 $\rightarrow$ Tape: `10000` (State O).
- Move head to start $\rightarrow$ Halt (State F).